

LESSON 6 : PERCENTAGES II (C)

Checkpoint 1

Simple Interest :

$$I = P \times R\% \times T$$

$$A = I + P$$

$$A = (P \times R\% \times T) + P \text{ or } P(1 + R\% \times T)$$

P = Principal

R = Rate

T = Time (by year)

I = Interest

A = Amount

1. Find the simplest interest of the following.

- | | | |
|--------------------------|--------------------------|-------------------|
| (a) Principal: \$2 000; | Interest rate 4% p.a.; | Period: 3 years |
| (b) Principal: \$300; | Interest rate 3.5% p.a.; | Period: 1.5 years |
| (c) Principal: \$60 000; | Interest rate 2% p.a.; | Period: 8 months |

- | | |
|----|--|
| 1. | $\$2\,000 \times 4\% \times 3 = \240 |
| 2. | $\$300 \times 3.5\% \times 1.5 = \15.75 |
| 3. | $\$60\,000 \times 2\% \times \frac{8}{12} = \800 |

Q1-4

Simple interest problem

2. Johnny deposits \$ 2 000 in Bank A at 10 % p.a., and \$ 3 500 in Bank B at 7 % p.a.

- (a) Find the simple interest he will receive in Bank A each year.
(b) Find the simple interest he will receive in Bank B each year.
(c) Find the total interest he will receive each year.

(a) Simple interest he will receive in Bank A each year +
 $= \$2000 \times 10\% \times 1$
 $= \underline{\underline{\$200}}$

(b) Simple interest he will receive in Bank B each year +
 $= \$3500 \times 7\% \times 1$
 $= \underline{\underline{\$245}}$

(c) Total interest he will receive each year
 $= \$ (200 + 245)$
 $= \underline{\underline{\$445}}$

3. Raymond deposits \$24 000 in a bank at a simple interest rate of 3% p.a. If he increases the deposit period from 2 years to 5 years, find the increase in the interest earned.

Increase in the interest earned +
 $= \$ (24\,000 \times 3\% \times 5 - 24\,000 \times 3\% \times 2)$
 $= \$24\,000 \times 3\% \times (5 - 2)$
 $= \underline{\underline{\$2160}}$

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4. Shirley deposits \$50 000 at a simple interest in a bank for 4 years. If the interest rate is increased from 4% to 9%, find the increase in the interest earned.

$$\begin{aligned}
 &\text{Increase in the interest earned} \leftarrow \\
 &= \$ (50\,000 \times 9\% \times 4 - 50\,000 \times 4\% \times 4) \\
 &= \$50\,000 \times 4 \times (9\% - 4\%) \\
 &= \underline{\underline{\$10\,000}}
 \end{aligned}$$

5. If \$8 000 is deposited in a bank at a simple interest rate of 2% p.a., how long will it take to receive an interest of \$960?

$$\begin{aligned}
 &\text{Let } t \text{ years be the required deposit period.} \leftarrow \\
 &\therefore I = P \times r\% \times t \leftarrow \\
 &\therefore 960 = 8000 \times 2\% \times t \\
 &0.02t = 0.12 \leftarrow \\
 &t = 6 \\
 &\therefore \text{The required deposit period is 6 years.} \leftarrow
 \end{aligned}$$

Q 5-8

Find rate/principal/time by Using
Simple interest formula

6. Sam borrows \$20 000 from a bank at a simple interest rate and he has to repay an amount of \$21 200 after 2 years. What is the annual interest rate?

$$\begin{aligned}
 &\text{Let } r\% \text{ be the annual interest rate.} \leftarrow \\
 &\therefore A = P(1 + r\% \times t) \leftarrow \\
 &\therefore 21\,200 = 20\,000 \times (1 + r\% \times 2) \\
 &1 + r\% \times 2 = 1.06 \\
 &r = 3 \\
 &\therefore \text{The interest rate is 3\% p.a.} \leftarrow
 \end{aligned}$$

7. If \$32 000 is deposited in a bank at a simple interest rate of 3% p.a., how long will it take to receive an amount of \$34 880?

$$\begin{aligned}
 &\text{Let } t \text{ years be the required deposit period.} \leftarrow \\
 &\therefore A = P(1 + r\% \times t) \leftarrow \\
 &\therefore 34\,880 = 32\,000 \times (1 + 3\% \times t) \\
 &1 + 3\% \times t = 1.09 \leftarrow \\
 &0.03t = 0.09 \\
 &t = 3 \\
 &\therefore \text{The required deposit period is 3 years.} \leftarrow
 \end{aligned}$$

8. Alex borrows \$ P from a bank at a simple interest rate 5% and he has to repay an amount of \$ 19550 after 3 years. Find the value of P .

$$\begin{aligned}
 19550 &= P(1 + 5\% \times 3) \\
 P &= \frac{19550}{1.15} \\
 P &= 17000
 \end{aligned}$$

9. A sum of \$12 000 is deposited at a simple interest rate of 3% p.a. For each of the following deposit periods, find

- (i) the simple interest,
(ii) the amount.

- (a) 2.5 years
(b) 18 months

Hints: 18 months = ? year

(a) (i) Simple interest = $\$12\,000 \times 3\% \times 2.5$
 $= \$900$
 (ii) Amount = $\$(12\,000 + 900)$
 $= \$12\,900$

(b) (i) Simple interest = $\$12\,000 \times 3\% \times \frac{18}{12}$
 $= \$540$
 (ii) Amount = $\$(12\,000 + 540)$
 $= \$12\,540$

10. Mrs. Yeung deposits some money in a bank at an interest rate of 3% p.a.. She gets \$600 simple interest for the first quarter of the year.

- (a) How much does Mrs. Yeung deposit at the beginning?
(b) If the interest rate is increased by 3 % p.a., find the simple interest that she will get for the remaining 3 quarters of the year.

(a) Let \$P be the amount that Mrs. Yeung deposits at the beginning.
 $P \times 3\% \times \frac{1}{4} = 600$
 $P = 80000$
 $\therefore$ Mrs. Yeung deposits \$80000 at the beginning.

(b) Simple interest that she will get
 $= \$80000 \times 6\% \times \frac{3}{4}$
 $= \$3600$

11. If the interest rate is 12% p.a. and the simple interest repaid after 15 months is \$1 500, find the principal borrowed.

Let \$P be the principal borrowed.
 $\therefore I = P \times r\% \times t$
 $\therefore 1500 = P \times 12\% \times \frac{15}{12}$
 $0.15P = 1500$
 $P = 10\,000$
 $\therefore$ The principal borrowed is \$10 000.

12. Maggie deposits \$15 000 in Bank A at a simple interest rate of 6% p.a. for 1.5 years, then she deposits the amount in Bank B at a simple interest rate of 3% p.a. for 2 years. Find

- (a) the amount she obtained from Bank A,
(b) the amount she obtained from Bank B,
(c) the total interest earned from both banks.

(a) Amount obtained from Bank A
 $= \$15\,000 \times (1 + 6\% \times 1.5)$
 $= \$16\,350$

(b) Amount obtained from Bank B
 $= \$16\,350 \times (1 + 3\% \times 2)$
 $= \$17\,331$

(c) Total interest earned from both banks
 $= \$17\,331 - \$15\,000$
 $= \$2331$

Checkpoint 2

Compound Interest :

$$A = I + P$$

$$A = P\left(1 + \frac{R\%}{n}\right)^t$$

$$I = P\left(1 + \frac{R\%}{n}\right)^t - P$$

P = Principal

R = Rate

n = number of Times (by year)

t = number of times during the period

I = Interest

A = Amount

13. \$10 000 is invested at 10% p.a. for 3 years. If the interest is compounded yearly, find

- (a) the amount,
(b) the compound interest after 3 years.

$$\begin{aligned} \text{(a) Amount} &= \$10\,000 \times (1 + 10\%)^3 \\ &= \underline{\underline{\$13\,310}} \end{aligned}$$

$$\begin{aligned} \text{(b) Compound interest} &= \$ (13\,310 - 10\,000) \\ &= \underline{\underline{\$3\,310}} \end{aligned}$$

Q 13-21

Compound interest problem

14. Tracy deposits \$30 000 in a bank at 3% p.a. compounded yearly, find the compound interest after 4 years.

$$\begin{aligned} \text{Compound interest} &= \$ [30\,000 \times (1 + 3\%)^4 - 30\,000] \\ &= \underline{\underline{\$3\,770}} \text{ (cor. to 3 sig. fig.)} \end{aligned}$$

15. Mr Tam invests \$8 000 for 3 years at 12% p.a. Find the amount he will obtain if the interest is compounded

- (a) half-yearly,
(b) quarterly,
(c) monthly.

$$\begin{aligned} \text{(a) The amount Mr Tam will obtain} &= \$8000 \times \left(1 + \frac{12\%}{2}\right)^{3 \times 2} \\ &= \underline{\underline{\$11\,300}} \text{ (cor. to 3 sig. fig.)} \end{aligned}$$

$$\begin{aligned} \text{(b) The amount Mr Tam will obtain} &= \$8000 \times \left(1 + \frac{12\%}{4}\right)^{3 \times 4} \\ &= \underline{\underline{\$11\,400}} \text{ (cor. to 3 sig. fig.)} \end{aligned}$$

$$\begin{aligned} \text{(c) The amount Mr Tam will obtain} &= \$8000 \times \left(1 + \frac{12\%}{12}\right)^{3 \times 12} \\ &= \underline{\underline{\$11\,400}} \text{ (cor. to 3 sig. fig.)} \end{aligned}$$

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16. Kate deposits \$ P in a bank at 6% p.a. and the interest is compounded half-yearly. If she obtains an amount of \$23 880 after 3 years, find

- the value of P ,
- the compound interest earned.

$$\begin{aligned} \text{(a)} \quad 23\,880 &= P \times \left(1 + \frac{6\%}{2}\right)^{3 \times 2} \\ 23\,880 &= P \times 1.03^6 \\ P &= 23\,880 \div 1.03^6 \\ &\approx 19\,999.1 \\ &= \underline{\underline{20\,000}} \text{ (cor. to 3 sig. fig.)} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \text{Compound interest} & \\ &\approx \$ (23\,880 - 19\,999.1) \\ &= \underline{\underline{\$3880}} \text{ (cor. to 3 sig. fig.)} \end{aligned}$$

17. Steve deposits a sum of money in a bank at 8% p.a. and the interest is compounded yearly. If he obtains an amount of \$99 144 after 2 years,

- find the principal,
- If the annual interest rate is unchanged and the interest is compounded monthly, find compound interest he will obtain after 9 months.

$$\begin{aligned} \text{(a)} \quad \text{Let } \$P \text{ be the principal.} & \\ \therefore 99\,144 &= P \times (1 + 8\%)^2 \\ 99\,144 &= P \times 1.08^2 \\ P &= 99\,144 \div 1.08^2 \\ &= 85\,000 \\ \therefore \text{The principal is } \$85\,000. & \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \text{Compound interest} & \\ &= \$ \left[85\,000 \times \left(1 + \frac{8\%}{12}\right)^9 - 85\,000 \right] \\ &= \underline{\underline{\$5240}} \text{ (cor. to 3 sig. fig.)} \end{aligned}$$

◆ 18. Alice deposits \$30 000 individually in two banks A and B . Bank A offers an interest rate of 6% p.a. compounded yearly, while bank B offers the same annual interest rate compounded quarterly. Assume that both deposits are for 3 years. Find

- the amount she obtained from bank A ,
- the amount she obtained from bank B ,
- the difference in the amounts she obtained from the two banks.

$$\begin{aligned} \text{(a)} \quad \text{Amount obtained from bank } A & \\ &= \$30\,000 \times (1 + 6\%)^3 \\ &= \underline{\underline{\$35\,730.48}} \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad \text{Difference in the amount} & \\ &\approx \$ (35\,868.55 - 35\,730.48) \\ &= \underline{\underline{\$138}} \text{ (cor. to 3 sig. fig.)} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \text{Amount obtained from bank } B & \\ &= \$30\,000 \times \left(1 + \frac{6\%}{4}\right)^{3 \times 4} \\ &\approx \$35\,868.55 \\ &= \underline{\underline{\$35\,900}} \text{ (cor. to 3 sig. fig.)} \end{aligned}$$

PRE S3 MATHS L6 – PERCENTAGES II (C)

19. A principal of \$50 000 is deposited for 2 years at an interest rate of 6% p.a.. Find the amount if it is compounded:

- (a) half-yearly;
 - (b) quarterly;
 - (c) monthly;
 - (d) daily (assume there are 365 days in a year).
- (Correct your answers to the nearest dollar.)

$$\begin{aligned} \text{(a) Amount} &= \$50,000 \left(1 + \frac{6\%}{2}\right)^{2 \times 2} = \$56,275 \\ \text{(b) Amount} &= \$50,000 \left(1 + \frac{6\%}{4}\right)^{2 \times 4} = \$56,325 \\ \text{(c) Amount} &= \$50,000 \left(1 + \frac{6\%}{12}\right)^{2 \times 12} = \$56,358 \\ \text{(d) Amount} &= \$50,000 \left(1 + \frac{6\%}{365}\right)^{2 \times 365} = \$56,374 \end{aligned}$$

20. Daniel deposits \$48 000 in a bank for 2 years and the interest is compounded monthly. If the interest rate is increased from 4% to 5%, find the increase in the amount obtained.

$$\begin{aligned} &\text{Increase in the amount} \quad \hookleftarrow \\ &= \$ \left[48\,000 \times \left(1 + \frac{5\%}{12}\right)^{2 \times 12} - 48\,000 \times \left(1 + \frac{4\%}{12}\right)^{2 \times 12} \right] \\ &= \$ \left[48\,000 \times \left(\frac{241}{240}\right)^{24} - 48\,000 \times \left(\frac{301}{300}\right)^{24} \right] \\ &= \underline{\underline{\$1050}} \quad (\text{cor. to 3 sig. fig.}) \end{aligned}$$

- ◆ 21. Chris deposits \$5 000 in bank *A* at 10% p.a. compounded half-yearly for 2 years, then he deposits the amount in bank *B* at a simple interest rate of 5% p.a. for $1\frac{1}{2}$ years. Find

- (a) the amount he obtained from bank *A*,
- (b) the total interest earned from two banks.

$$\begin{aligned} \text{(a) Amount Chris obtained from bank } A & \\ &= \$5000 \times \left(1 + \frac{10\%}{2}\right)^{2 \times 2} \\ &\approx \$6077.53 \quad \hookleftarrow \\ &= \underline{\underline{\$6080}} \quad (\text{cor. to 3 sig. fig.}) \end{aligned}$$

$$\begin{aligned} \text{(b) Simple interest earned from bank } B & \quad \hookleftarrow \\ &\approx \$6077.53 \times 5\% \times 1\frac{1}{2} \quad \hookleftarrow \\ &\approx \$455.81 \\ \text{Total interest earned in two banks} & \quad \hookleftarrow \\ &\approx \$[(6077.53 - 5000) + 455.81] \\ &= \underline{\underline{\$1530}} \quad (\text{cor. to 3 sig. fig.}) \quad \hookleftarrow \end{aligned}$$

22. \$3 000 is deposited for 5 years at an interest rate of 10% p.a., find the different between the simple interest and the compound interest compounded yearly.

$$\begin{aligned} \text{Compound interest after 5 years} &= \$3\,000 \times (1 + 10\%)^5 - \$3\,000 \\ &= \$1\,831.53 \\ \text{Simple interest after 5 years} &= \$3\,000 \times 10\% \times 5 \\ &= \$1\,500 \\ \therefore \text{The difference} &= \$1\,831.53 - \$1\,500 \\ &= \$331.53 \end{aligned}$$

Checkpoint 3

A man borrows \$ 5 000 from a bank at 12 % p.a., compounded monthly. He repays \$ 500 at the end of each month. Find how much he still owes the bank after the second repayment, correct to 1 decimal place.

May		June		July	
At the beginning e.g. 1/5/2020	\$5000 <i>is borrowed from the bank</i>	Then he repays \$500 on 1/6/2020	After repayment, \$5050 – 500 = 4550	Then he repays \$500 on 1/6/2020	After repayment, \$4595.5 – 500 = 4095.5
31/5/2020 And before repay the money	Total money $\$5000 \left(1 + \frac{12\%}{12}\right)^{(12)(1/12)}$ = \$5050 <i>borrowed from the bank</i>	30/6/2020 And before repay the money	Total money $\$4550 \left(1 + \frac{12\%}{12}\right)^{(12)(1/12)}$ = \$4595.5 <i>borrowed from the bank</i>		<i>borrowed from the bank</i>

23. A man borrows \$20 000 from a bank at 10% p.a. and repays \$5 000 at the end of each year. Find the amount he still owes after the fourth payment.

After 1st payment, amount he owes = $\$20,000(1 + 10\%) - \$5,000 = \$17,000$;
 After 2nd payment, amount he owes = $\$17,000(1 + 10\%) - \$5,000 = \$13,700$;
 After 3rd payment, amount he owes = $\$13,700(1 + 10\%) - \$5,000 = \$10,070$;
 $\therefore$ After 4th payment, amount he owes = $\$10,070(1 + 10\%) - \$5,000 = \$6,077$.

Q 23-25
Repay money problem

24. A man borrows \$5 000 from a bank at 12% p.a., compounded monthly. He repays \$500 at the end of each month. Find how much he still owes the bank after the second repayment, correct to 1 decimal place.

After 1 month, he owes the bank by
 $5000(1 + 1\%)^1 - 500 = \4550
 After 2 months, he owes the bank by
 $4550(1 + 1\%) - 500 = \$4095.5$

25. Jason borrowed \$35 000 from a bank at an interest rate of 18% p.a. compounded monthly, and he had to repay \$10 000 at the end of each month.

- (a) How much, correct to the nearest dollar, did he owe the bank at the beginning of the third month?
 (b) How many repayments were needed to settle the loan?

(a) At beginning of 2nd month, amount he owed = $\$35,000(1 + \frac{18\%}{12}) - \$10,000 = \$25,525$,
 $\therefore$ At beginning of 3rd month, amount he owed = $\$25,525(1 + 1.5\%) - \$10,000 = \$15,908$

(b) At beginning of 4th month, amount he owed = $\$15,908(1 + 1.5\%) - \$10,000$
 = \$6,146, at end of 4th month, amount he owed = $\$6,146(1 + 1.5\%)$
 = \$6,239 < \$10,000. *Ans. 4 payments were needed.*

Checkpoint 4

Miss Kwan's total annual income for the year 2019/20 is \$ 300 000. If her salaries tax allowance is \$ 120 000, how much salaries tax should she pay

Step 1 Covert total annual income into Net chargeable income

$$\begin{aligned}\text{Net chargeable income} &= 300\,000 - 120\,000 \\ &= \$ 180\,000\end{aligned}$$

Step 2 How many \$ 50 000 can be divided from
Net chargeable income?

ANS: 3 and the remainder is \$ 30 000

Step 3 Calculation

$\therefore$ Miss Kwan's salaries tax payable

$$\begin{aligned}&= 50\,000 \times 2\% + 50\,000 \times 6\% + 50\,000 \times 10\% + 30\,000 \times 14\% \\ &= \$ 13\,200\end{aligned}$$

Net chargeable income	Tax rate	Salary tax
On the first \$ 50 000	2 %	\$ 1 000
On the next \$ 50 000	6 %	\$ 3 000
On the next \$ 50 000	10 %	\$ 5 000
On the next \$ 50 000	14 %	\$ 7 000
Remainder	17 %	

26. The net chargeable income of Thomas is \$155 000. How much salaries tax should he pay?

$$\text{Net taxable income} = \$ 155\,000$$

$$= \$ (50\,000 + 50\,000 + 50\,000 + 5\,000)$$

$$\begin{aligned}\text{Salaries tax} &= \$ 50\,000 \times 2\% + \$ 50\,000 \times 6\% + \$ 50\,000 \times 10\% + \$ 5\,000 \times 14\% \\ &= \$ 9\,700\end{aligned}$$

27. If the basic allowance for a single person is \$108 000, find the amount of salaries tax for each of the following monthly income.

(a) \$10 000

(b) \$63 000

$$(a) \text{ Net taxable income} = \$ 10\,000 \times 12 - \$108\,000 = \$ 12\,000$$

$$\text{Salaries tax} = \$12\,000 \times 2\% = \$ 240$$

$$(b) \text{ Net taxable income} = \$ 63\,000 \times 12 - \$108\,000 = \$ 648\,000$$

$$= \$ (50\,000 + 50\,000 + 50\,000 + 50\,000 + 448\,000)$$

$$\begin{aligned}\text{Salaries tax} &= \$ 50\,000 \times (2\% + 6\% + 10\% + 14\%) + \$ 448\,000 \times 17\% \\ &= \$ 16\,000 + 76\,160 \\ &= \$ 92\,160\end{aligned}$$

28. Miss Lo's salaries tax payable charged at progressive rates is \$240 for this year. If she has an allowance of \$216 000, find her annual income.

Let \$x be the net chargeable income of Miss Lo.

$$x \times 2\% = 240$$

$$x = 12\,000$$

$\therefore$ The net chargeable income of Miss Lo is \$12 000.

$$\begin{aligned}\text{Annual income} &= \$216\,000 + \$12\,000 \\ &= \underline{\underline{\$228\,000}}\end{aligned}$$